CASE STUDY · AGENTIC AI

Progressive Authentication for Agentic AI

We built a login model that lets anonymous shoppers try Agentic AI before they sign in. The goal was to protect the platform while still growing the number of recognized users.

Role: Product Designer, end-to-end Surface: Agentic AI (Ask Blue)

WHAT WE WERE SOLVING

The problem

Only about 23% of BestBuy.com sessions are recognized. Because Agentic AI required a login before the first interaction, most anonymous shoppers never even got to try it. Leadership wanted a way to open the experience to more people without exposing the platform to bots, runaway infrastructure costs, or conversations that disappear the moment a tab closes.

~23%
of sessions were recognized before this work

~79%
of sessions were anonymous and unable to use it

THE CONSTRAINTS

What made this hard

Authentication is a sensitive moment. Ask too early and people leave before they see the value. Wait too long and the platform takes on real risk, from rising AI costs and bot abuse to conversations that disappear when a tab closes. The experience had to support three identity states: visitor, recognized, and authenticated. It also had to stay consistent across multiple tabs and clearly separate optional sign-in prompts from actions that truly require an account, such as viewing order history or saving to a wishlist.

Three identity states
Visitor (VT), recognized (UT), and authenticated (AT) states all need to behave predictably. Conversations should carry over as people move between them.

Cross-tab conversations
Each browser tab has its own conversation under the same visitor identity. The experience needs to remain consistent whether someone opens one tab or five.

Soft nudge vs. hard gate
Optional prompts and required logins need to look and behave differently, so people always know whether they can continue without signing in.

Cost & abuse guardrails
Conversation limits and rate limits keep infrastructure costs under control and discourage bots without getting in the way of real shoppers.

MY ROLE

What I did

I led the progressive authentication experience for Agentic AI from problem framing through prototyping and handoff. I worked with product and engineering to decide when the prompt should appear and how to explain the benefit without making sign-in feel forced.

- Defined how anonymous (VT), recognized (UT), and authenticated (AT) users move between states while keeping conversations intact.
- Worked with product and engineering to define the engagement threshold and feature flag that trigger the optional sign-in prompt.
- Prototyped embedded cards, floating cards, popovers, and drawers for four moments: general use, viewing history, saving an item, and ending a session.
- Wrote and refined copy around the benefit someone had just experienced instead of falling back on a generic sign-in message.
- Set frequency caps and dismissal rules so the prompt would not keep interrupting the conversation.
- Designed a drawer for actions that require an account, including conversation history.

PROCESS

How we worked

Discovery → Design → Prototype → Align. I started from the PRD's identity model, mapping every trigger point before drawing a single screen — then pressure-tested direction through two prototyping rounds before engineering handoff.

01
Discovery
Mapped the PRD identity model from visitor to recognized to authenticated, along with every point where sign-in could be introduced.

02
Design
Explored nudge placement and tone across embedded cards, floating cards, popovers, and drawers.

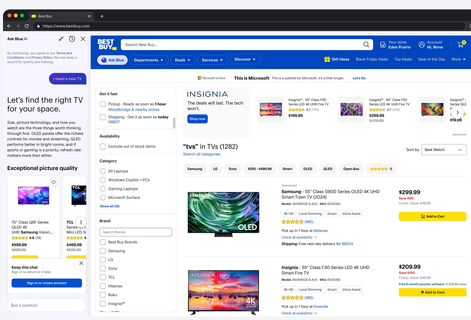
03
Prototype
Used two prototype rounds to refine the copy, timing, and dismissal behavior.

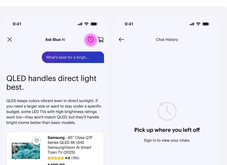
04
Align
Worked with product and engineering to agree on trigger logic, feature flags, and edge cases.

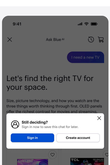
WHAT IT LOOKS LIKE

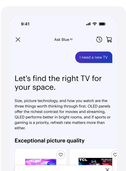
Screens

One prompt pattern, adapted to four different moments in the conversation.









PRINCIPLES

Key design decisions

Six practical principles shaped the timing, placement, and wording of each prompt.

01
Value before the ask
The prompt appears only after 3+ conversation turns and a product recommendation, once the shopper has had a chance to see the assistant's value.

02
Skipping must feel safe
Choosing not to sign in never interrupts or restarts the conversation. Anonymous shoppers can simply keep going.

03
One ask per session
The prompt appears once per session. If someone dismisses it, we respect that choice and do not ask again.

04
Right surface for the moment
An inline card works during the conversation, a popover suits lighter moments, and a drawer is reserved for account-required actions like history.

05
Benefit-driven copy
The message reflects what the shopper can keep or gain, such as saved finds, free shipping, or order tracking, instead of using a generic sign-in request.

06
Configurable by design
The turn threshold, prompt cap, and visitor retention window can all be adjusted with feature flags after launch.

Success metrics & status

WHAT SUCCESS LOOKS LIKE

After two rounds of prototyping, the work is now in engineering handoff. Product and analytics will track these measures after launch:

Recognized sessions
Increase the share of anonymous sessions that become recognized, signed-in sessions.

Agentic AI engagement
Increase conversation depth and the number of people who return to use the assistant.

Add-to-cart from AI
Increase add-to-cart activity that starts within an Agentic AI conversation.